Decision Trees

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Decision Trees before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Types of Classification Error

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Training and Testing

Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Operational Labels

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Decision Trees trace and record the first step you can explain without help.

Use W2: Verify the candidate scores for **Fitting** a **decision tree**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Decision Trees explanation against the supplied facts before discussing it.

Use the worked case for **Fitting** a **decision tree**. Proposed explanation: “A near **branch** must predict blocked”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Fit a new tree independently

New **training** rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree.

Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree **prediction**s and compare accuracy with the **training**-majority baseline.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

Connect the Decision Trees target to a trace and a justified conclusion.

Build a tree for science books: first ask science?; then, for yes, pages<=100? Trace science/100 and history/70. What should happen to science with missing pages?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Decision Trees case and show the deciding evidence.

First ask animal book? If no, Other. If yes, ask pages<100; yes gives Short Animal, no gives Longer Animal.

Trace 90-page animal and science books.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A tree asks topic is space, then pages at most 60. A space book has 70 pages. Trace both answers and explain why you cannot stop at the first question.

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Decision Trees adds to Transparent Book Recommender and keep the evidence.

Create a transparent tree for hard constraints before applying preference scores.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Build a tree for science books: first ask science?; then, for yes, pages<=100? Trace science/100 and history/70. What should happen to science with missing pages?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Name the parts and decisions

Name the parts and decisions. Apply the supplied Decision Trees facts and keep your reasoning.

Identify the **root**, its two **branch**es, **leaf prediction**s and **depth** of the worked fitted tree. Which parts were selected using **training** labels?

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Verify the candidate scores

> Verify the candidate scores. Apply the supplied Decision Trees facts and keep your reasoning.

For each near **branch** and each tall **branch**, count blocked and clear labels and choose a **leaf**. Give the total errors for each candidate question.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Separate fitting from prediction

> Separate fitting from prediction. Apply the supplied Decision Trees facts and keep your reasoning.

A learner sees a blocked validation case with near=0 and changes that **leaf** to blocked before reporting the score. Explain what changed and how to report the original fitted tree fairly.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Fit a new tree independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Fit a new tree independently. Apply the supplied Decision Trees facts and keep your reasoning.

New **training** rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree.

Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree **prediction**s and compare accuracy with the **training**-majority baseline.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Audit a tie and a claim

> Audit a tie and a claim. Apply the supplied Decision Trees facts and keep your reasoning.

If near and tall make the same **training** error count, which question does this code retain and why? Does retaining that question prove it will be better on future trips?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Decision Trees

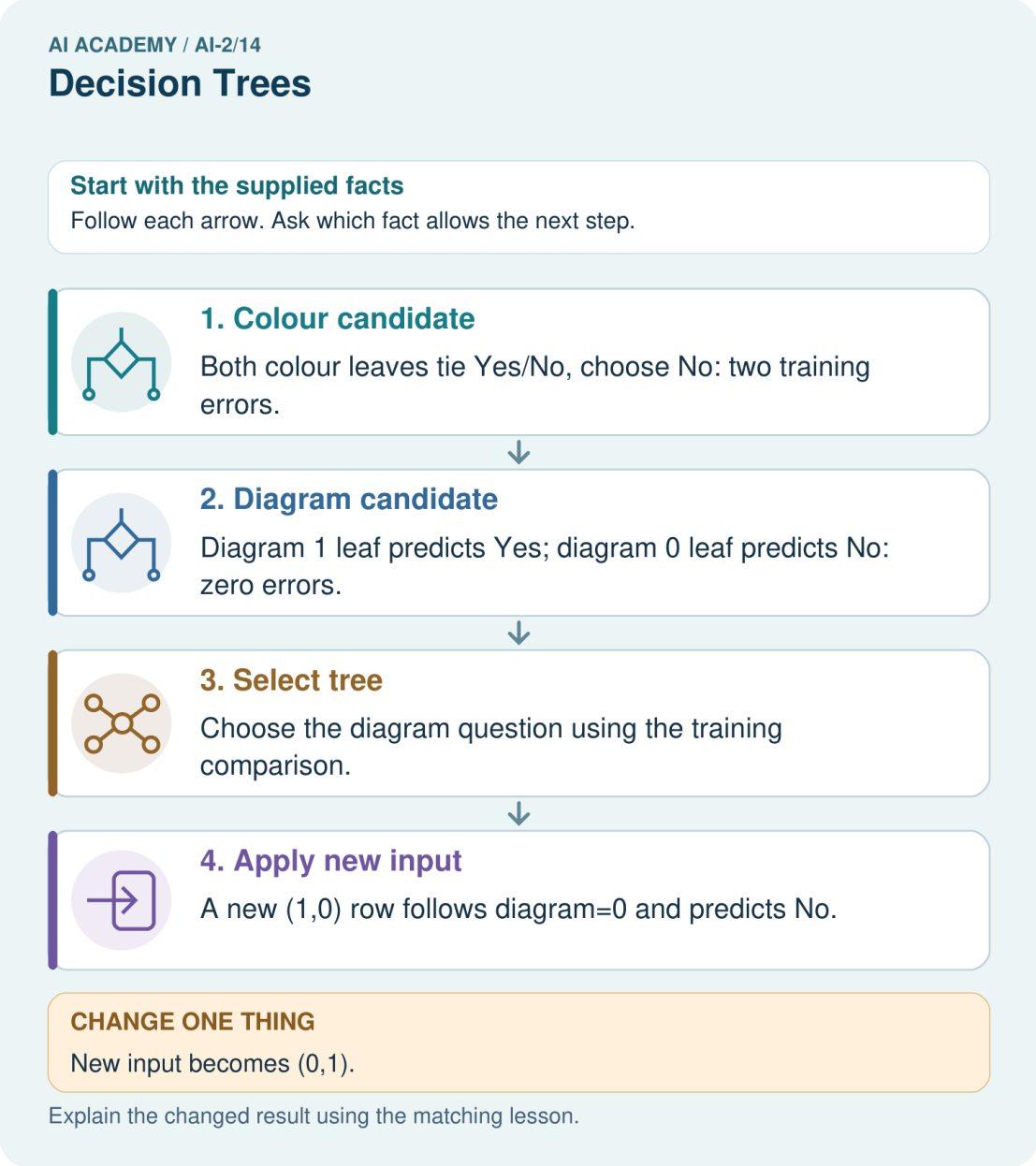
> A fictional toy classifier predicts whether a book needs an illustration label. Explain the

> labelled step before checking the changed condition.

A fictional toy classifier predicts whether a book needs an illustration label.

Training rows (colour_cover,diagram_inside,label): (1,1,Yes),(1,0,No),(0,1,Yes),(0,0,No).

Candidates ask colour_cover or diagram_inside; each **leaf** uses its **training** majority. A **leaf** tie chooses No. Select fewer **training** errors.



Follow the labelled picture with the matching explanation. Open the vector diagram

Why does changing colour alone not change this tree's **prediction**?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Fit or hand-specify recommender classification rules transparently, retaining candidate definitions and **training**-only selection **evidence**.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Fit or hand-specify recommender classification rules transparently, retaining candidate definitions and **training**-only selection **evidence**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional toy classifier predicts whether a book needs an illustration label.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? A new (1,0) row follows diagram=0 and predicts No.

1. Select tree

2. Diagram candidate

3. Colour candidate

4. Apply new **input**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Apply new **input** A new (1,0) row follows diagram=0 and predicts No. A question becomes a fitted tree only after its leaves and selection procedure are specified.

The **training** comparison prefers the feature that separates these labels. **Prediction** follows the selected question; it does not reselect the tree using a new case's answer.

Save candidate leaves and errors so the choice can be repeated.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: New **input** becomes (0,1). Which conclusion is supported by the supplied facts?

1. The fitted tree predicts Yes.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The fitted tree predicts Yes. It follows diagram=1; cover colour is not the selected question.

Repair a missing building block

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?.

Return to this week and explain what you changed.

Review **Training** and Testing (ai-2/11). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan counts?.

Return to this week and explain what you changed.

Review Operational Labels (ai-2/8). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Why can we not resolve the disagreement merely by taking a majority of reviewers?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

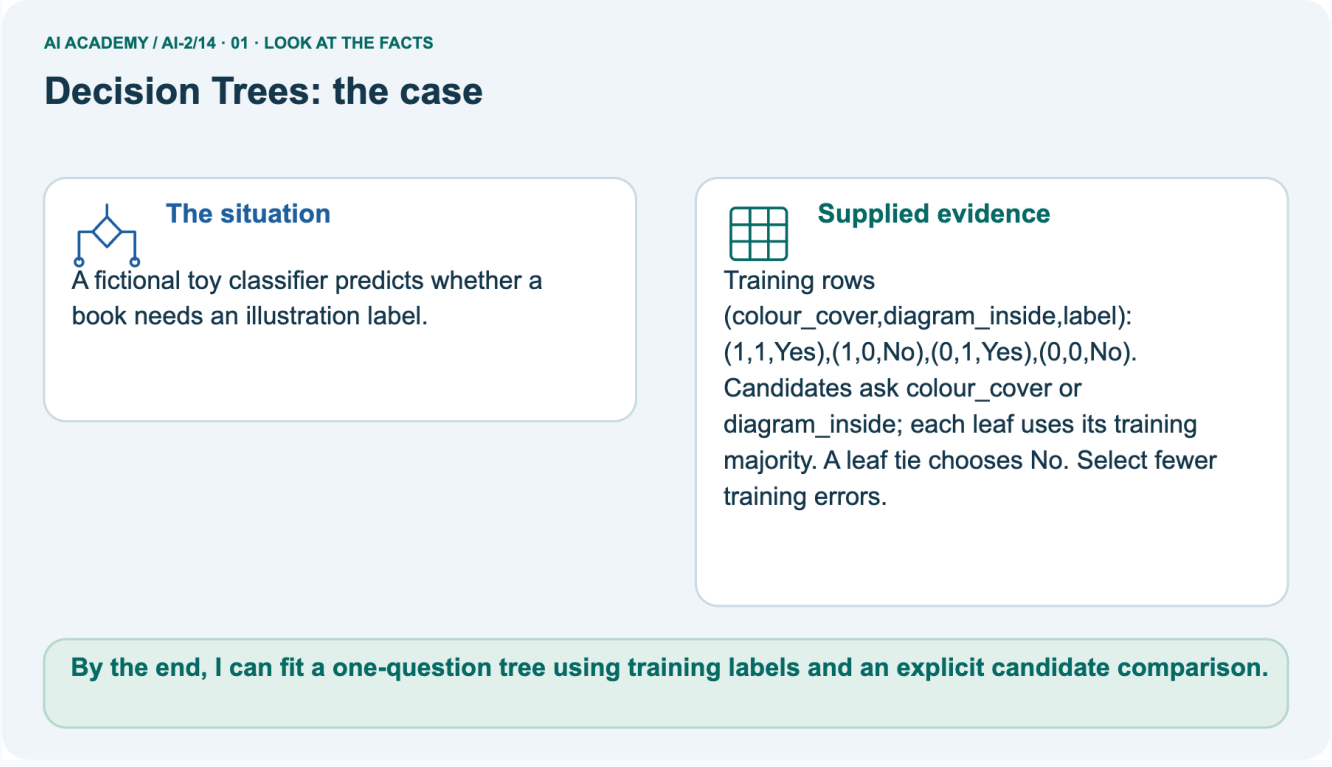
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Decision Trees picture from its labels, then record a relevant limit.

01. Decision Trees: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

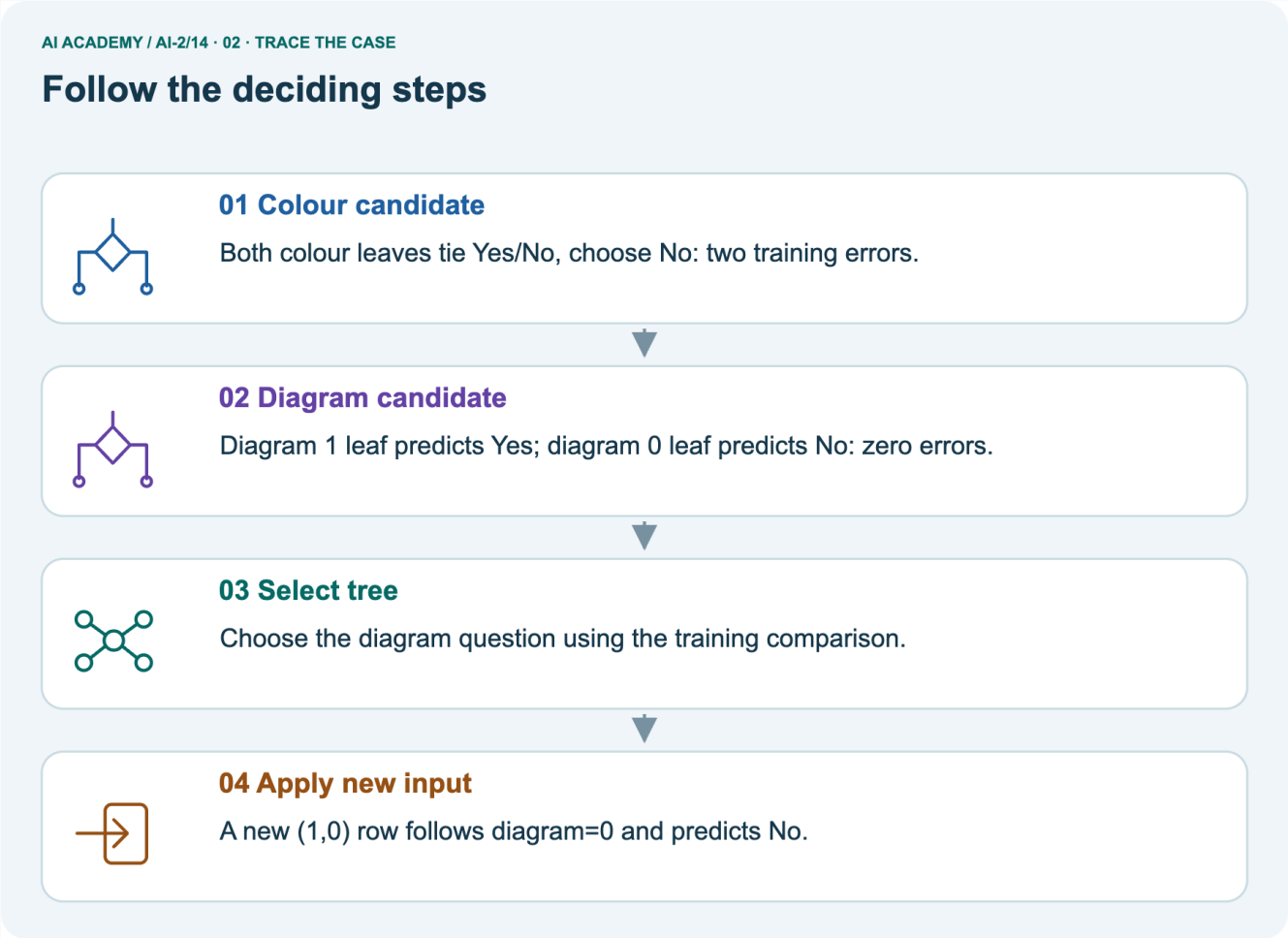
A fictional toy classifier predicts whether a book needs an illustration label. Supplied facts: **Training** rows (colour_cover,diagram_inside,label): (1,1,Yes),(1,0,No),(0,1,Yes),(0,0,No).

Candidates ask colour_cover or diagram_inside; each **leaf** uses its **training** majority. A **leaf** tie chooses No. Select fewer **training** errors.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



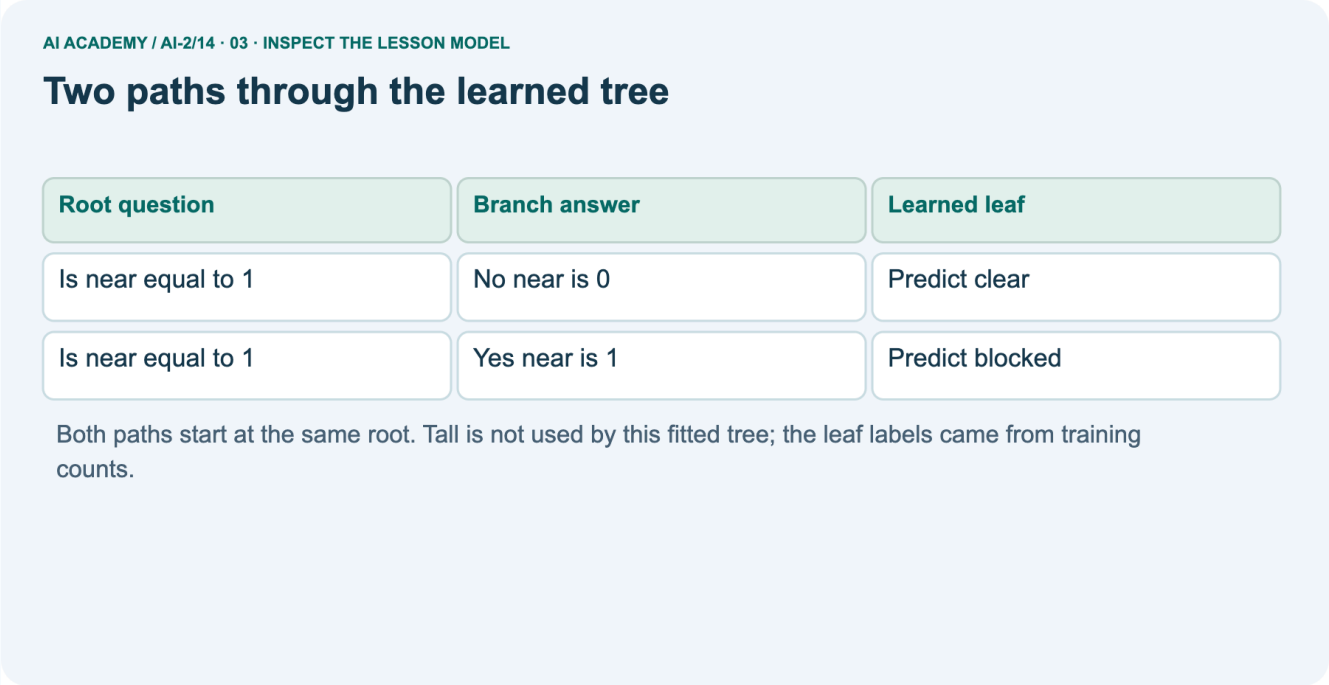
Follow the labelled picture with the matching explanation. Open the vector diagram

Colour candidate: Both colour leaves tie Yes/No, choose No: two **training** errors. → Diagram candidate: Diagram 1 **leaf** predicts Yes; diagram 0 **leaf** predicts No: zero errors. → Select tree: Choose the diagram question using the **training** comparison. → Apply new **input**: A new (1,0) row follows diagram=0 and predicts No.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two paths through the learned tree



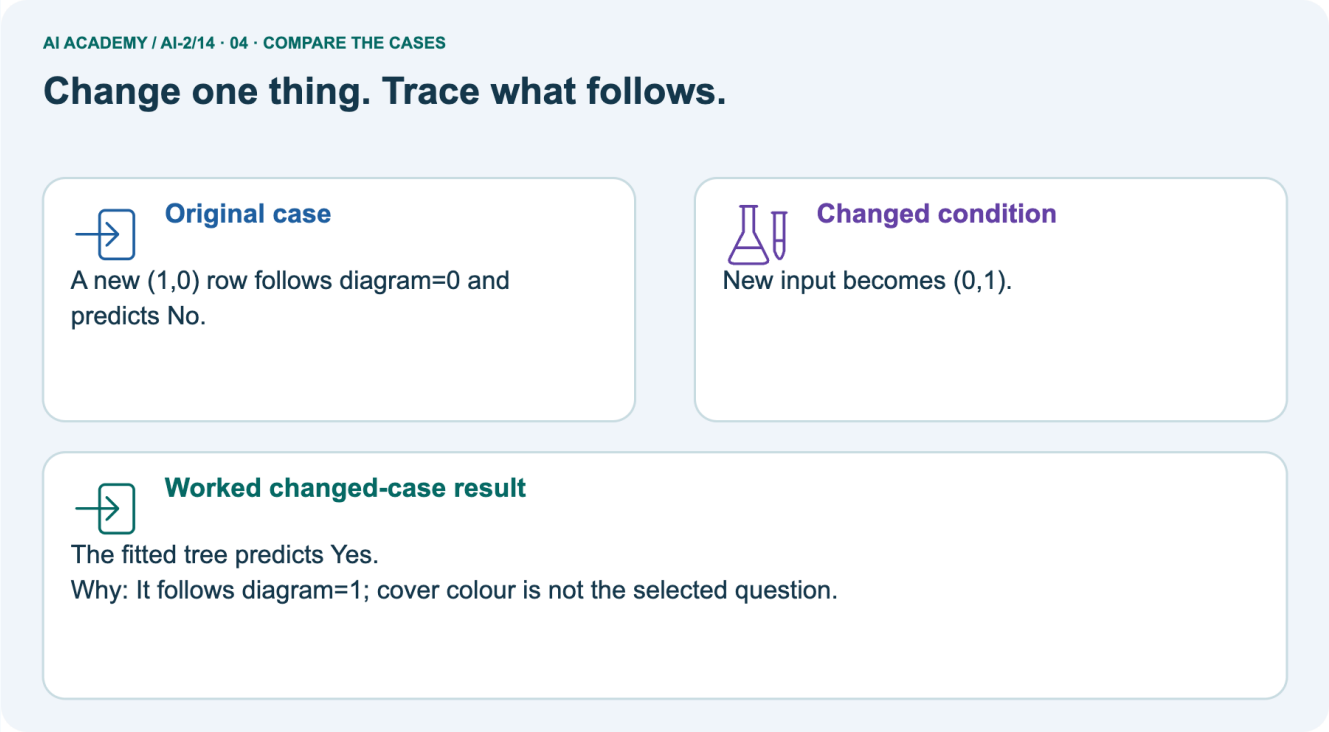
Follow the labelled picture with the matching explanation. Open the vector diagram

Two paths through the learned tree. Is near equal to 1 | No near is 0 | Predict clear. Is near equal to 1 | Yes near is 1 | Predict blocked

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

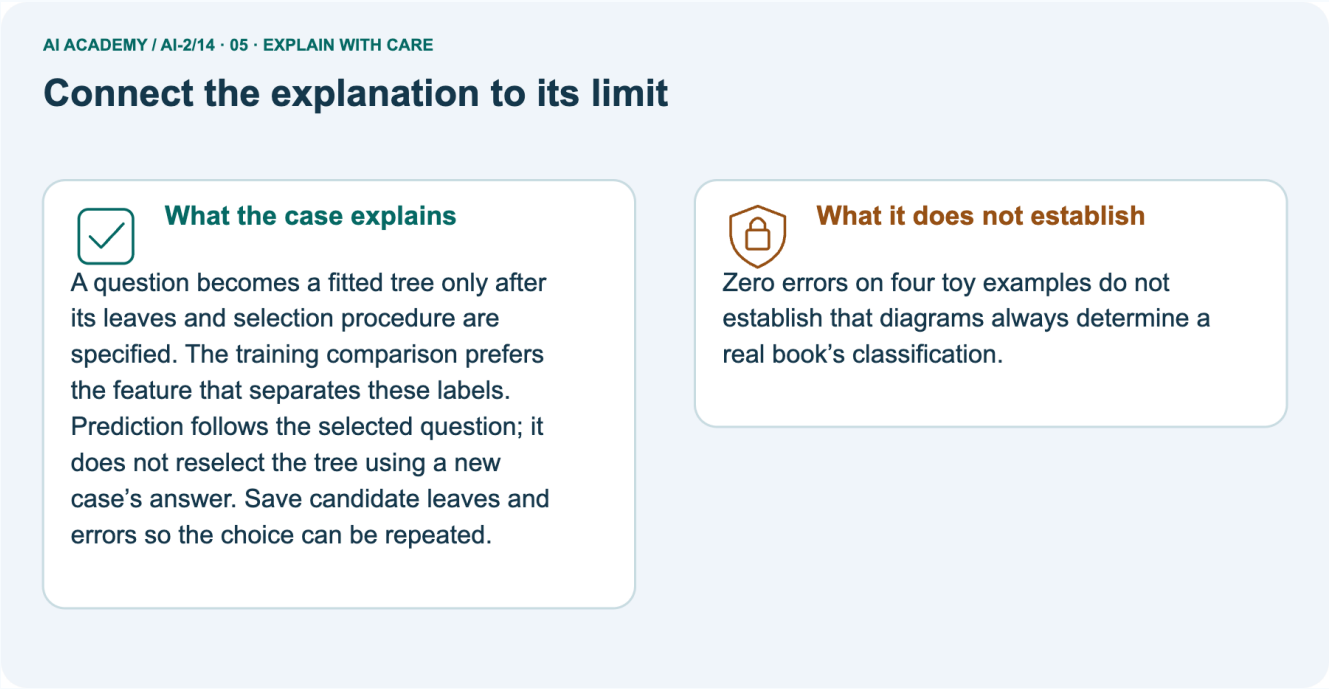
Original case: A new (1,0) row follows diagram=0 and predicts No. Changed condition: New **input** becomes (0,1). Expected result: The fitted tree predicts Yes.

Reason: It follows diagram=1; cover colour is not the selected question.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A question becomes a fitted tree only after its leaves and selection procedure are specified. The **training** comparison prefers the feature that separates these labels.

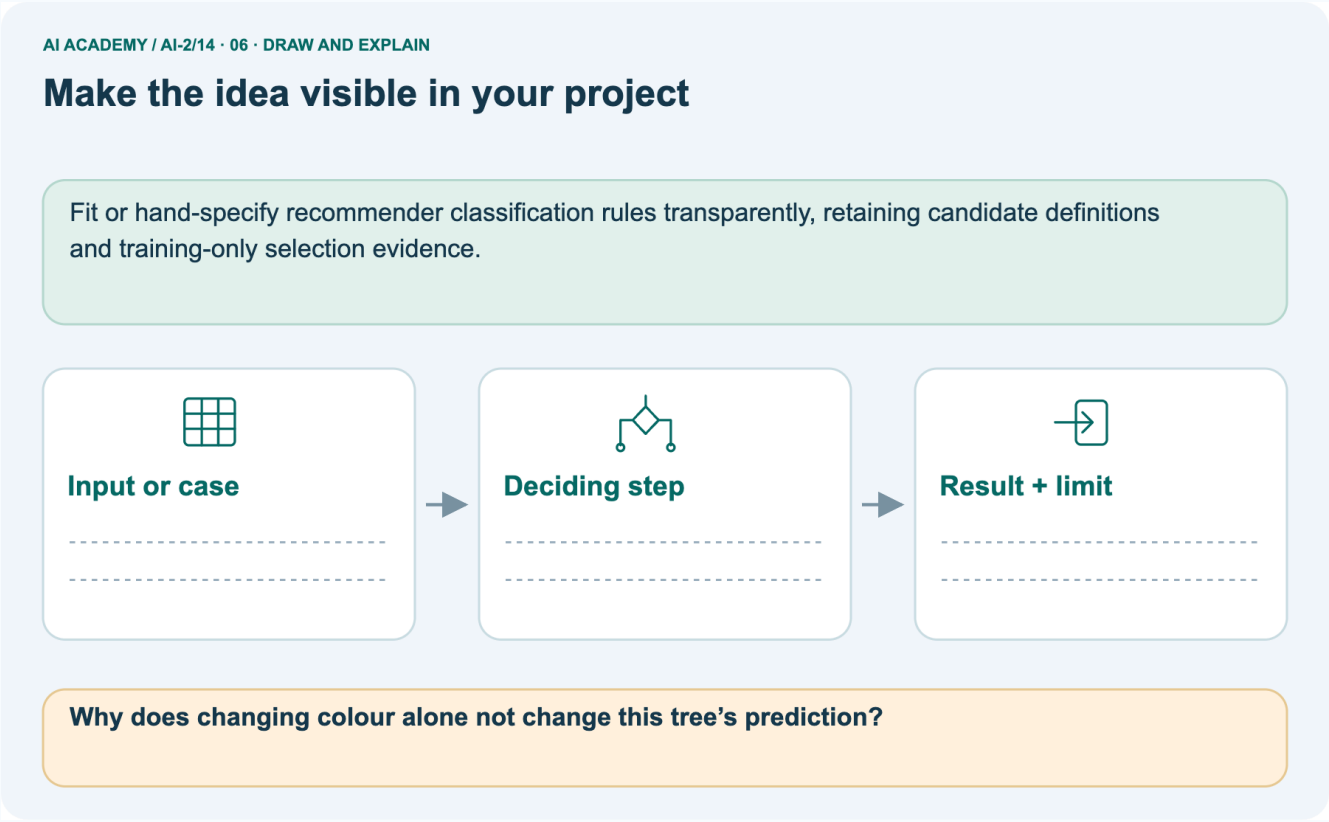
Prediction follows the selected question; it does not reselect the tree using a new case's answer. Save candidate leaves and errors so the choice can be repeated.

Limit: Zero errors on four toy examples do not establish that diagrams always determine a real book's classification.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Fit or hand-specify recommender classification rules transparently, retaining candidate definitions and **training**-only selection **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: Why does changing colour alone not change this tree's **prediction**?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.